HEALTHCARE ANALYTICS

Hospital Performance Intelligence

Patient demand service workload and payer coverage

An executive review of where hospital management should focus when patient volume, service intensity and financial coverage point to different priorities.



Management priority

Separate coverage follow-up by payer context and assess service workload before reallocating capacity. Use the report to identify the population, evidence and owner for each investigation.

Analysis period January 2011 to February 2022
Prepared October 2026

EXECUTIVE REPORT NAVIGATION

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Decision focus

The report connects each business question to a finding, a management implication and a proposed action. Dashboard pages provide the visual evidence. Metric definitions, calculations and data-quality rules are included in Section VI for closer review.

All dashboard screenshots show the overall All filter context. Annual charts end in 2021; overall totals include the partial 2022 period.

BUSINESS CONTEXT AND SCOPE

I. Project Overview

Hospital management needs to distinguish the number of people served from the work generated by their visits. Patient counts, encounter counts, procedure activity and recorded claim costs answer different operating questions. Bringing them into one report helps managers decide which service or population requires closer review.

Purpose

I developed Hospital Performance Intelligence to connect executive performance, utilization, payer coverage and patient behavior. The report was built in Power BI Desktop and published to Power BI Service, with consistent controls across four pages.

Objectives

The analysis identifies where demand is concentrated, compares procedure workload across services, locates coverage patterns that warrant financial review, and shows which patient cohorts and service classes generate repeat activity.

Coverage	Included in the analysis
Activity	27,891 encounters and 47,701 procedure records
Population	974 distinct patients with recorded encounters
Model	Encounters and procedures, with patient, payer, organization and date tables
Time	January 2, 2011 through February 6, 2022; annual trends cover 2011 to 2021
Report views	Executive Overview, Encounters & Utilization, Financial & Coverage, Patient Behavior

Business use

A manager can start with an overall signal, narrow the view by period or population, compare the relevant measures and assign the next investigation. The report supports workload, coverage and access discussions using a consistent analytical context.

Data context: This executive case study uses public synthetic hospital records published by Maven Analytics and sourced from SyntheticMass. The named hospital is part of the dataset context.

QUESTIONS THAT GUIDE THE ANALYSIS

II. Business Questions

Where is demand changing

How have encounter volume and patient reach changed over time? The distinction matters because a rise in encounters can reflect more patients, more repeat visits, or a different service mix. The annual charts provide a starting point for investigating the change.

Which services create the most workload

Which services contribute the most procedures in total, and which involve more procedures per encounter? Total volume helps locate the largest activity pool; intensity helps identify services where equal visit counts may represent different workloads.

Where should coverage follow up begin

Which payer groups account for zero recorded coverage, and how does their claim-cost profile differ? The answer helps finance separate self-pay support from insured eligibility or classification checks before reviewing payment records.

Which populations drive repeat demand

Which encounter-age cohorts generate the most visits, and which services have the most encounters per patient? These patterns help care operations frame a review of follow-up access, continuity and service coordination.

Business question	Primary report view	Decision owner
Overall demand and patient reach	Executive Overview	Management
Volume and procedure intensity	Encounters & Utilization	Operations
Payer costs and zero coverage	Financial & Coverage	Finance
Cohorts and repeat use	Patient Behavior	Care operations

Year, Encounter Class, Payer, Age Group, Gender and Month filters keep these questions specific. A finding should always be reviewed with its selected period and population.

FINDINGS AND MANAGEMENT IMPLICATIONS

III. Key Insights

974 patients generated 27,891 encounters

The full-period average is 28.64 encounters per patient. Patient reach alone therefore understates service activity. Planning discussions should examine repeat use within the relevant service and period.

Wellness has the highest procedure intensity

Wellness averages 4.81 procedures per encounter, compared with 1.71 overall. Ambulatory care has the largest total procedure volume at 17,822. These identify different priorities for workload review.

Two services account for 58.7 percent of claim costs

Ambulatory and Urgentcare contribute $36.28M and $23.35M respectively. Their combined cost concentration makes them a practical starting point for examining service and procedure mix.

Zero coverage is concentrated in No_Insurance

Of 13,586 zero-coverage encounters, 8,807 are in No_Insurance, representing 64.8% of the zero-coverage population. Finance should distinguish this group from insured encounters before selecting a follow-up workflow.

The 65 plus cohort accounts for 67.7 percent of encounters

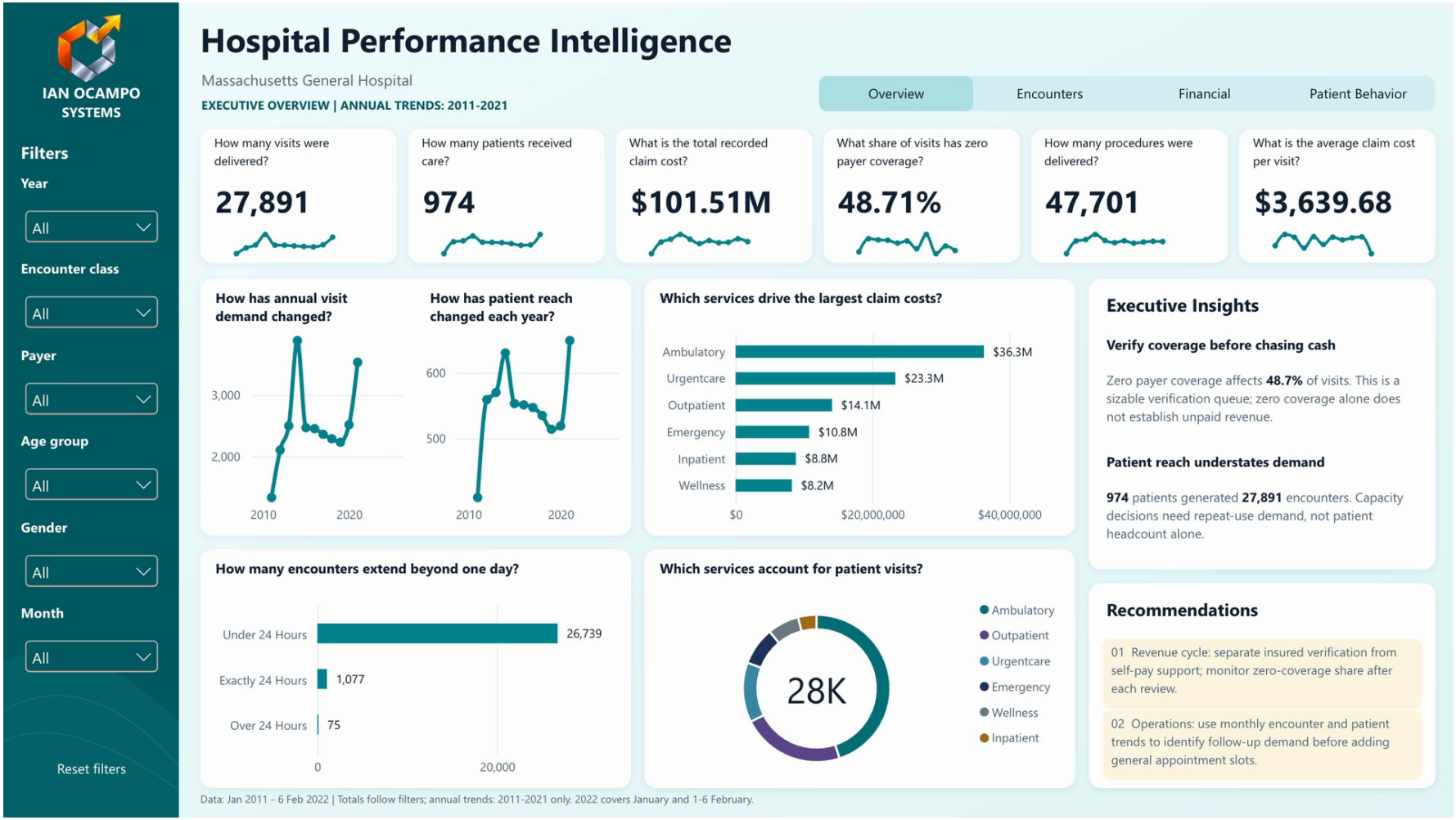
Patients aged 65 and above at the encounter generated 18,883 visits. This activity share supports a focused review of appointment access and continuity for that cohort.

2021 demand rose within a variable annual pattern

Encounters increased 40.2%, from 2,517 in 2020 to 3,530 in 2021. The complete-year peak remains 3,882 in 2014. Management should investigate period-specific changes rather than assume uninterrupted growth.

III. KEY INSIGHTS / DASHBOARD EVIDENCE 1 OF 4

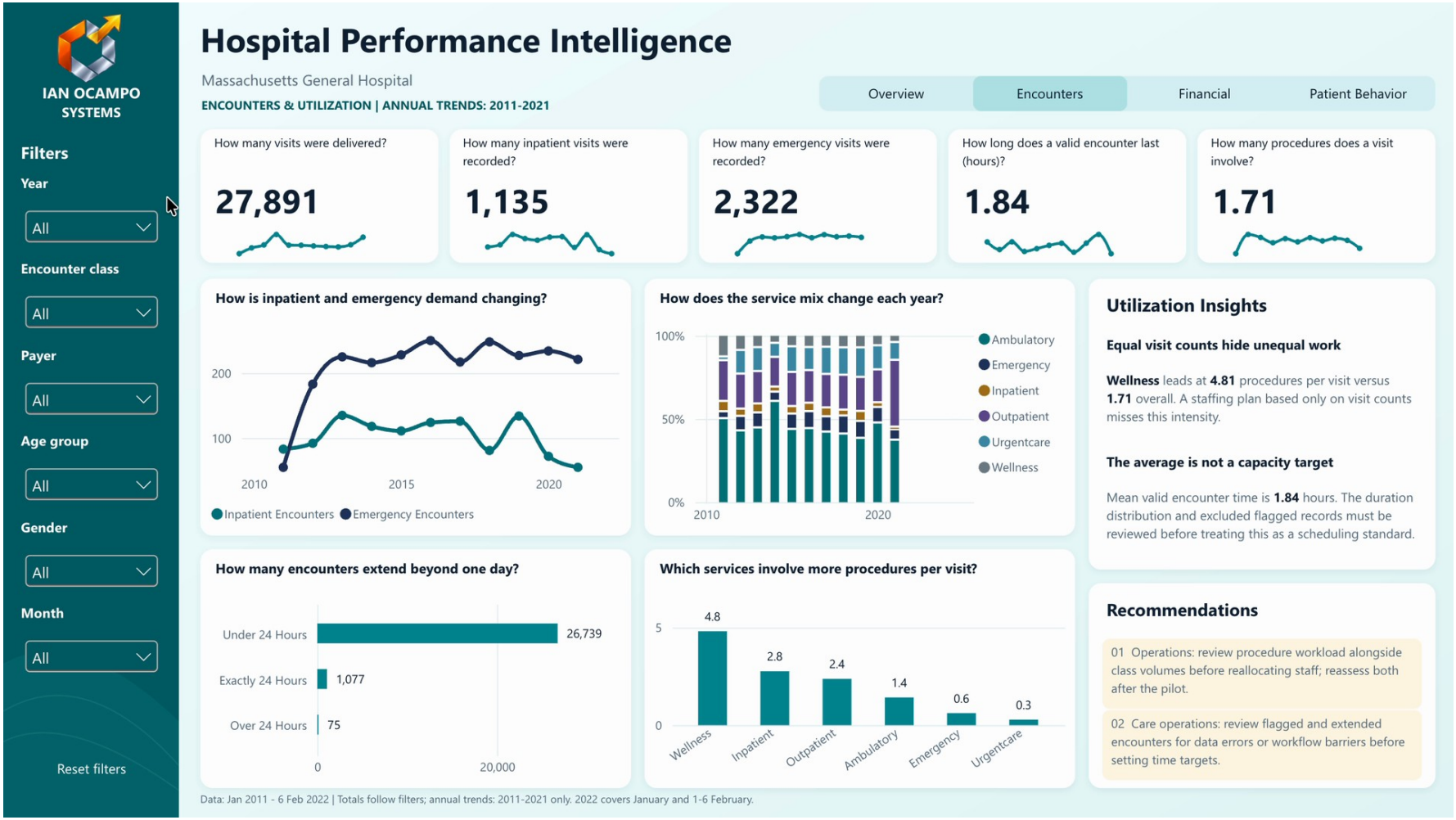
Executive Overview



Compare patient reach, visits, procedures, recorded costs and coverage in one view. The service and trend visuals help management choose the next area to investigate.

III. KEY INSIGHTS / DASHBOARD EVIDENCE 2 OF 4

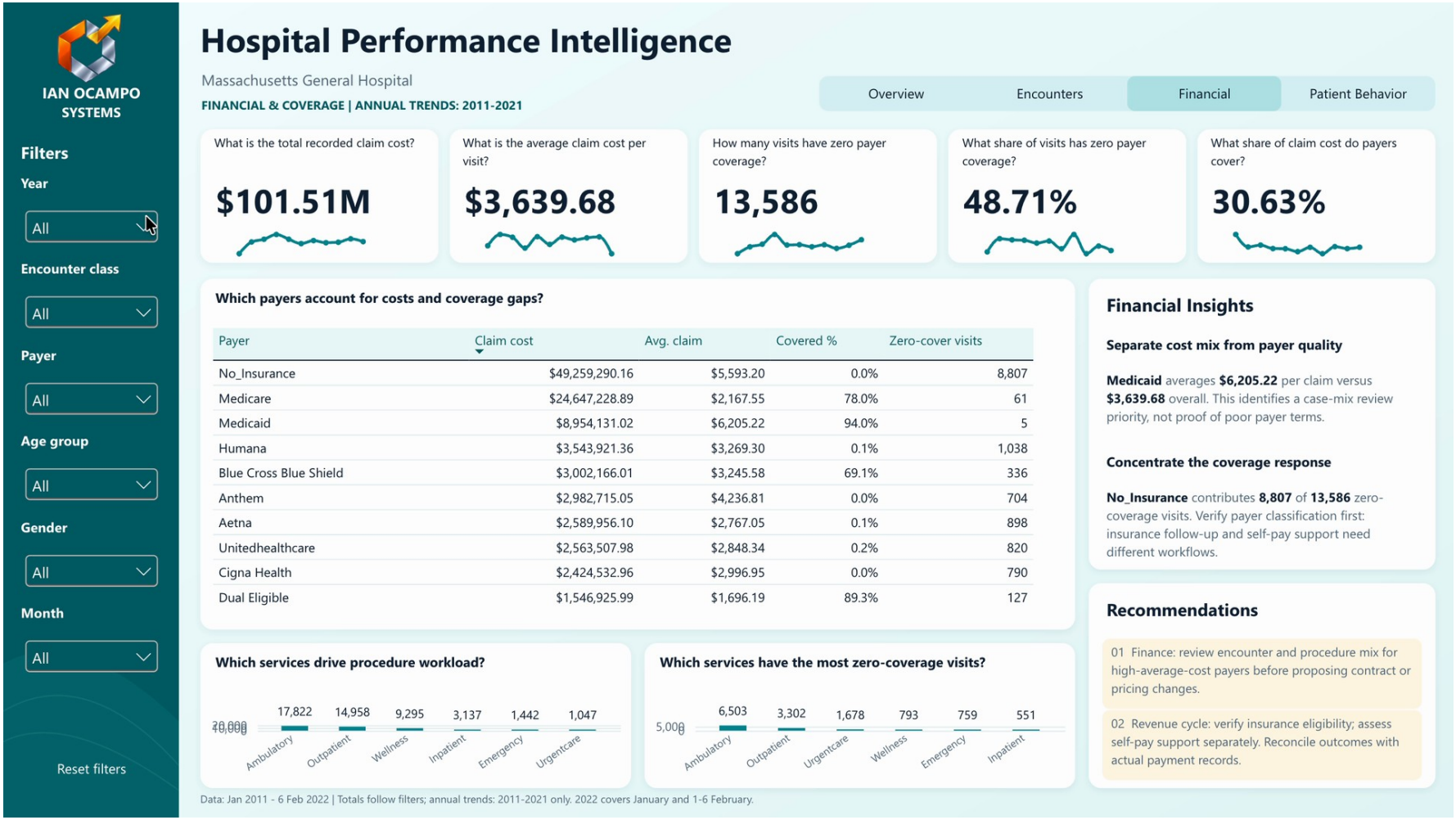
Encounters and Utilization



Compare encounter demand with procedure intensity and valid elapsed duration. Use service mix to focus a workload review before changing staffing or appointment capacity.

III. KEY INSIGHTS / DASHBOARD EVIDENCE 3 OF 4

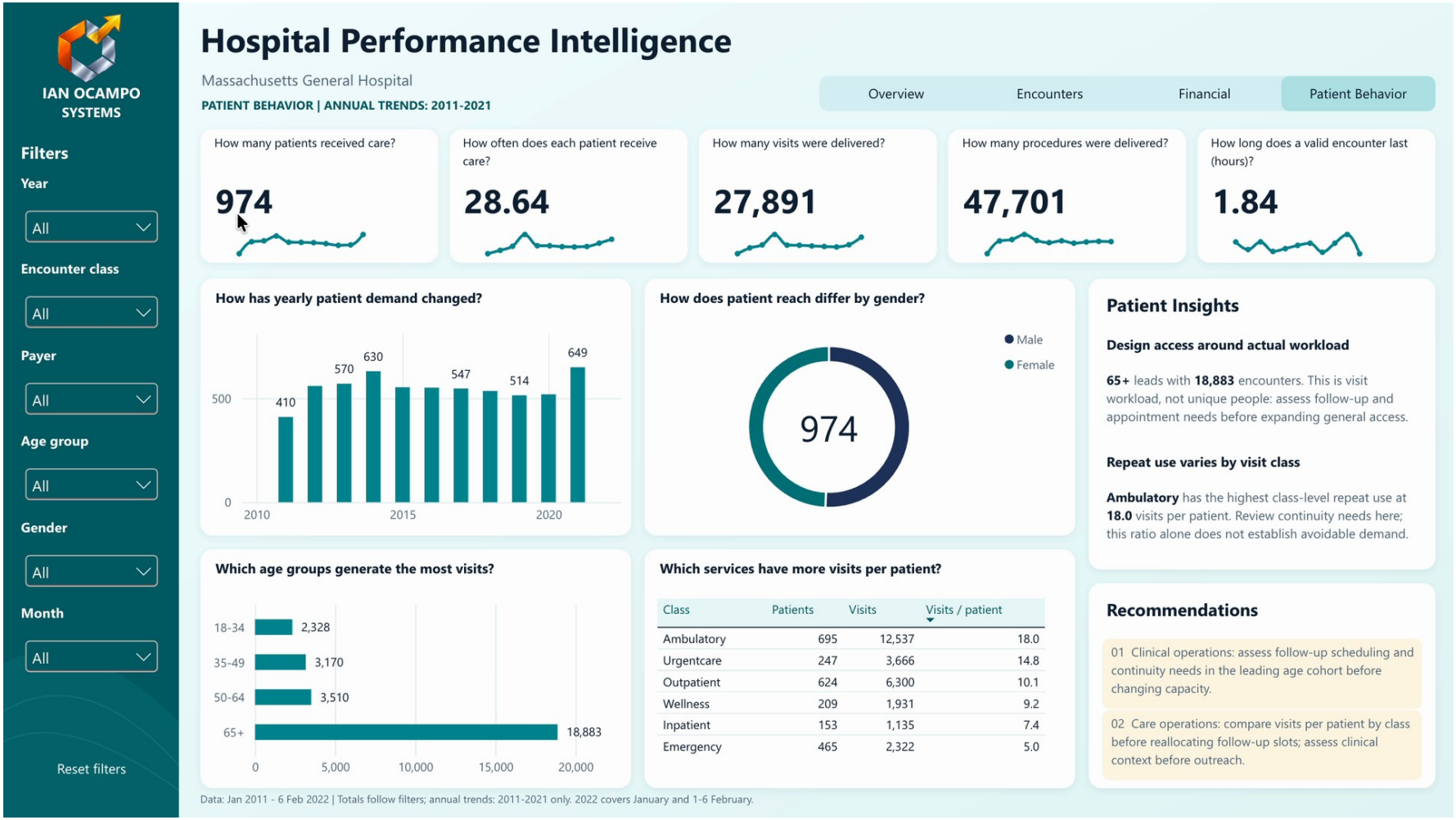
Financial and Coverage



Use payer-level detail to locate cost and coverage patterns. Recorded claim costs and payer coverage require reconciliation with eligibility and payment records before financial action.

III. KEY INSIGHTS / DASHBOARD EVIDENCE 4 OF 4

Patient Behavior



Separate distinct patient reach from repeat service demand. Use encounter-age cohorts and encounters per patient by service class to frame continuity and access reviews.

PROPOSED ACTIONS BASED ON THE FINDINGS

IV. Business Recommendations

Prioritize coverage verification

Finance or revenue-cycle teams should separate No_Insurance from insured zero-coverage encounters. Remove zero-cost records from an exposure review, then reconcile a sample against eligibility, payer classification and payment records. Track the reason confirmed for each reviewed case.

Review service workload before reallocating capacity

Operations should start with Ambulatory because of its total procedure volume and Wellness because of its procedure intensity. Compare procedure types, appointment duration and observed staff time before selecting a limited scheduling or staffing intervention.

Investigate continuity and appointment access

Care operations should examine the 65+ encounter-age cohort and services with high encounters per patient. Review follow-up availability and reasons for repeat use. Clinical context is required before treating repeat activity as potentially avoidable.

Resolve duration exceptions

The data owner should review the 13 records flagged over 30 days and confirm start and stop timestamps. Retain the agreed duration definition and an exception count in recurring data-quality checks.

Use a consistent management review

The report owner should record the business question, selected filters, finding, action owner and follow-up evidence at each review. Use matching periods and populations when assessing whether an intervention changed the operating result.

Priority sequence: establish reliable coverage and duration context, investigate the selected services and cohorts, then test and measure a focused intervention.

IV. BUSINESS RECOMMENDATIONS / SUGGESTED SEQUENCE

Implementation priorities

The proposed rollout separates validation from operational change. Owners and timing should be confirmed for the organization using the report.

Window	Suggested owner	Deliverable
Days 1 to 30	Finance and data owner	Agree metric definitions; review coverage and duration exceptions; document confirmed reasons.
Days 31 to 60	Operations and care operations	Review service workload and repeat-demand cohorts; identify one specific constraint and select a limited intervention.
Days 61 to 90	Service manager and report owner	Compare the intervention with a matched baseline, record the result and decide whether to extend or revise it.

Evidence required before a decision

Review area	Evidence to add	Decision it informs
Coverage	Eligibility, payer classification and payment records	Select the appropriate follow-up workflow.
Workload	Procedure mix, appointment duration and observed staff time	Assess the capacity constraint and proposed response.
Repeat demand	Reasons for visits, follow-up availability and care context	Assess continuity or access changes.
Duration	Confirmed encounter start and stop timestamps	Resolve exceptions and retain a defensible duration measure.

This sequence gives each proposed action a responsible team and a defined evidence requirement. Targets should be agreed after current operating data has been validated.

DELIVERED REPORTING CAPABILITY AND EXPECTED BUSINESS VALUE

V. Impact / Value

The delivered solution provides four connected report views, 14 defined DAX measures and six consistent filters on each page. Management can narrow a broad question to a specific period, service, payer or patient cohort while retaining a common calculation framework.

Delivered capability	Business value supported	How value would be measured
Connected service views	More focused workload and capacity investigations	Review time, constraints identified and results of tested scheduling changes.
Payer-level coverage detail	Separate insured follow-up from self-pay support	Cases reviewed, confirmed exception reasons and follow-up completion.
Patient cohort and repeat-use views	Better framed continuity and access reviews	Follow-up availability and repeat-use patterns in matched cohorts.
Explicit definitions and data checks	Consistent interpretation across reviews	Reconciliation differences and unresolved data exceptions.

Summary

The analysis locates the main investigation priorities: workload varies by service volume and intensity, zero coverage is concentrated in No_Insurance, and older encounter-age cohorts account for most activity. The report keeps these patterns visible without treating them as proof of their underlying causes.

Management conclusion

Begin with targeted verification and service reviews, then test a limited intervention using matched measures and populations. The case study demonstrates the reporting capability delivered. Financial savings, improved collections or changes in capacity would be outcomes to measure during implementation, rather than benefits already established by the historical analysis.

IMPLEMENTED REPORTING STACK AND ANALYTICAL METHODOLOGY

VI. Technology Stack and Technical Methodology

Component	Role in the solution
Power BI Desktop	Report authoring, model configuration and visual analysis.
Power Query	CSV import, typing, encounter-age enrichment and analytical flags.
DAX	Counts, distinct counts, sums, ratios and valid-duration averages.
Power BI Service	Published report access and interactive navigation.
CSV source tables	Encounter, procedure, patient, payer, organization and calendar inputs.

Preparation and relationships

Power Query promotes CSV headers and assigns text, date, numeric and currency types. Patient birthdates are joined to encounters to calculate age at the visit and ordered age bands. Additional flags identify zero coverage, inpatient encounters and emergency encounters.

The model uses active, single-direction relationships from patient, payer, organization and date dimensions to encounters, then from encounters to procedures. This preserves encounter context when comparing visits and linked procedure activity.

Analytical techniques

The report combines descriptive analysis, annual comparisons, service and cohort segmentation, and ratio analysis. Baseline totals were reconciled to the embedded model. Unique encounter identifiers, matching patient and payer keys, and procedure-to-encounter links were checked.

Analytical source: Supplied Power BI model and published report. Dataset: Maven Analytics Hospital Patient Records, sourced from SyntheticMass. https://mavenanalytics.io/data-playground/hospital-patient-records

VI. TECHNICAL METHODOLOGY / DAX MEASURES

Metric definitions and calculations

All-data baselines include the partial 2022 period. Measures respond to the active Power BI filter context. The table below defines all 14 measures and their baseline results.

Measure	Calculation	Baseline
Total Encounters	Count of encounter rows	27,891
Unique Patients	Distinct encounter Patient identifiers	974
Total Procedures	Count of linked procedure rows	47,701
Total Claim Cost	Sum of recorded claim costs	$101,514,375.52
Total Payer Coverage	Sum of recorded payer coverage	$31,097,506.99
Average Claim Cost	Total Claim Cost / Total Encounters	$3,639.68
Zero Coverage Encounters	Sum of zero-coverage flags	13,586
Zero Coverage %	Zero Coverage Encounters / Total Encounters	48.71%
Coverage %	Total Payer Coverage / Total Claim Cost	30.63%
Encounters per Patient	Total Encounters / Unique Patients	28.64
Procedures per Encounter	Total Procedures / Total Encounters	1.71
Average Duration Hours	Mean of ValidDurationHours	1.84 hours
Inpatient Encounters	Sum of inpatient flags	1,135
Emergency Encounters	Sum of emergency flags	2,322

DAX functions include COUNTROWS, DISTINCTCOUNT, SUM, DIVIDE and AVERAGE. Coverage % is cost-weighted. Encounters per Patient is a full-period average, not an annual rate.

VI. TECHNICAL METHODOLOGY / ANALYTICAL CONTEXT

Filters and data quality

Filter	Technical meaning and analytical use
Year	Filters calendar years. Annual trend visuals are restricted to 2011 through 2021.
Encounter Class	Selects one or more service categories for a comparable workload and cost review.
Payer	Filters coverage groups across encounter activity and linked procedures.
Age Group	Uses age at the encounter, based on the patient birthdate and visit date.
Gender	Segments recorded demographic context while retaining the selected service and period.
Month	Filters calendar months. Combine with Year for a specific period; with All years selected, it pools that month across years.

Duration quality rule

Average Duration Hours uses ValidDurationHours and excludes 13 records flagged over 30 days. This changes the baseline mean from 7.27 to 1.84 hours. Those records remain in encounter and cost totals. Elapsed encounter duration is not a staffing target or bed occupancy measure.

Coverage definition

Zero payer coverage is recorded on 13,586 encounters, including 89 zero-cost records. There are therefore 13,497 positive-cost encounters with zero recorded payer coverage. Actual unpaid balances require separate payment evidence.

Time and count controls

Overall totals run through February 6, 2022, including 222 encounters in that partial year. Annual charts end in 2021. Distinct patient counts across service classes or years must not be summed because the same person can appear in more than one group.